

II. BREAST IMAGING LEXICON — MAMMOGRAPHY

Table 1. BI-RADS® Mammography Lexicon Overview

Breast Tissue		Terms
Breast density ¹	(A) The breasts are almost entirely fatty	
	(B) There are scattered areas of fibroglandular density	
	(C) The breasts are heterogeneously dense, which may obscure small masses	
	(D) The breasts are extremely dense, which lowers the sensitivity of mammography	
Findings		Terms
A. Masses	1. Shape	a. Oval b. Lobulated c. Round d. Irregular
	2. Margin	a. Circumscribed b. Obscured c. Indistinct d. Spiculated
	3. Density	a. Fat-containing b. Low density c. Equal density d. High density
B. Calcifications	1. Typically benign	a. Skin b. Vascular c. Coarse d. Large rod-like e. Round f. Rim g. Layering h. Suture
	2. Suspicious morphology	a. Amorphous b. Coarse heterogeneous c. Fine pleomorphic d. Fine linear or fine linear-branching
	3. Distribution	a. Diffuse b. Regional c. Grouped d. Linear e. Segmental
C. Architectural distortion		
D. Asymmetries	1. Global asymmetry 2. Asymmetry 3. Focal asymmetry	

E. Lymph nodes	1. Intramammary	a. Normal b. Abnormal
	2. Axillary	a. Normal b. Abnormal
F. Skin lesions		
G. Dilated ducts	1. Multiple	
	2. Solitary	
H. Associated features	1. Skin retraction	
	2. Nipple retraction	
	3. Skin thickening	
	4. Trabecular thickening	
I. Special cases	1. Gynecomastia	
	2. Implants and other forms of augmentation	
	3. Mastectomy	
J. Location of finding	1. Laterality	
	2. Location in breast	
	3. Depth and/or distance from the nipple	

¹In July 2025, the FDA approved an alternative standard that would allow the interpreting physician to provide an overall assessment of breast density that is phrased in the singular phrasing. The singular phrasing is intended to be used in reports of unilateral mammograms. Please see the Guidance section for more information.

III. BREAST DENSITY

Breast density is defined as the relative proportion of breast parenchyma to fat within the breast. Mammographically, breast parenchyma is radiodense relative to radiolucent fat. There are four categories of breast density (Table 2): the breasts are almost entirely fatty, there are scattered areas of fibroglandular density, the breasts are heterogeneously dense which may obscure small masses, and the breasts are extremely dense which limits the sensitivity of mammography. Reporting breast density is required by Food and Drug Administration (FDA) legislation as of September 2024. Language included in the report both to the referring physician and the patient must follow the federal guidance.

People with heterogeneously dense or extremely dense breasts are considered to have dense breasts, while those with almost entirely fatty breasts or scattered areas of fibroglandular density are considered to be non-dense. Most commonly, the interpreting radiologist subjectively categorizes the breast density using visual inspection of the breast parenchyma on the craniocaudal and mediolateral DM or SM views, although FDA-approved automated density measurement software is available. With the BI-RADS 5th Edition, the classification of breast density changed from an evaluation of the percentage of dense tissue to an assessment based on the likelihood that overlying tissue may obscure a cancer. This classification continues in v2025. If tissue in one region is dense and potentially masking an underlying cancer, then the breast density classification should be based on an assessment of this region, rather than the overall amount of parenchymal tissue. If each breast falls into a different density category, the greater density of the two is assigned to that examination. Breast density can be variable from year to year, depending on multiple factors such as weight gain or loss, estrogen replacement therapy, menopause, etc. Therefore, breast density is updated at each examination to accurately reflect the current breast density.

Breast density is significant for two reasons: mammographic accuracy and breast cancer risk. The impact of breast density on mammographic accuracy is due to the masking effect of dense fibroglandular tissue. With film-screen mammography, sensitivity declines significantly from fatty breasts to extremely dense breasts [9]. The masking effect caused by dense breast tissue has been reduced by advances in technology, first with the adoption of digital mammography, and, more recently, with DBT. In the DMIST trial, the sensitivity of mammography in women with dense breasts increased from 55% with film mammography to 70% with digital mammography [10]. With DBT, the interpreting radiologist can scroll through images of the breast slice by slice, further reducing the masking effect of dense breast tissue. In women with dense breasts, the cancer detection rate increased from 4.5 per 1000 women screened with digital mammography to 5.8 per 1000 women screened with DBT ($p < 0.001$) [11].

As with increasing age, pathogenic genetic variants, family history, and reproductive factors, having dense breasts confers an increased inherent risk for developing breast cancer. Compared to the breasts of women with scattered fibroglandular tissue, the odds ratio of breast cancer is 0.6 for almost entirely fatty, 1.4 for heterogeneously dense, and 1.6 for extremely dense breasts [12].

Table 2. Breast Density Categories¹

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|--|
| (A) The breasts are almost entirely fatty |
| (B) There are scattered areas of fibroglandular density |
| (C) The breasts are heterogeneously dense, which may obscure small masses |
| (D) The breasts are extremely dense, which lowers the sensitivity of mammography |

¹In July 2025, the FDA approved an alternative standard that would allow the interpreting physician to provide an overall assessment of breast density that is phrased in the singular phrasing. The singular phrasing is intended to be used in reports of unilateral mammograms. Please see the Guidance section for more information.

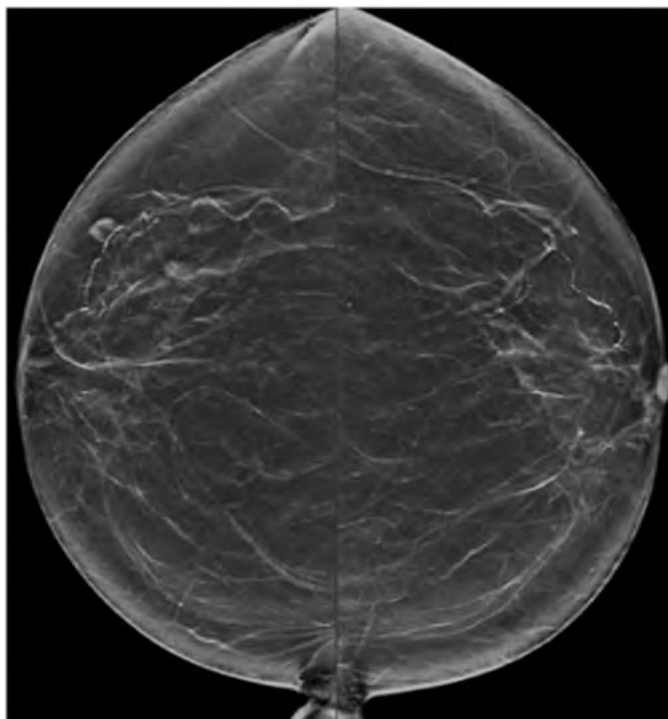


Figure 1 – Breast Density: BI-RADS breast composition category A, almost entirely fatty (SM).

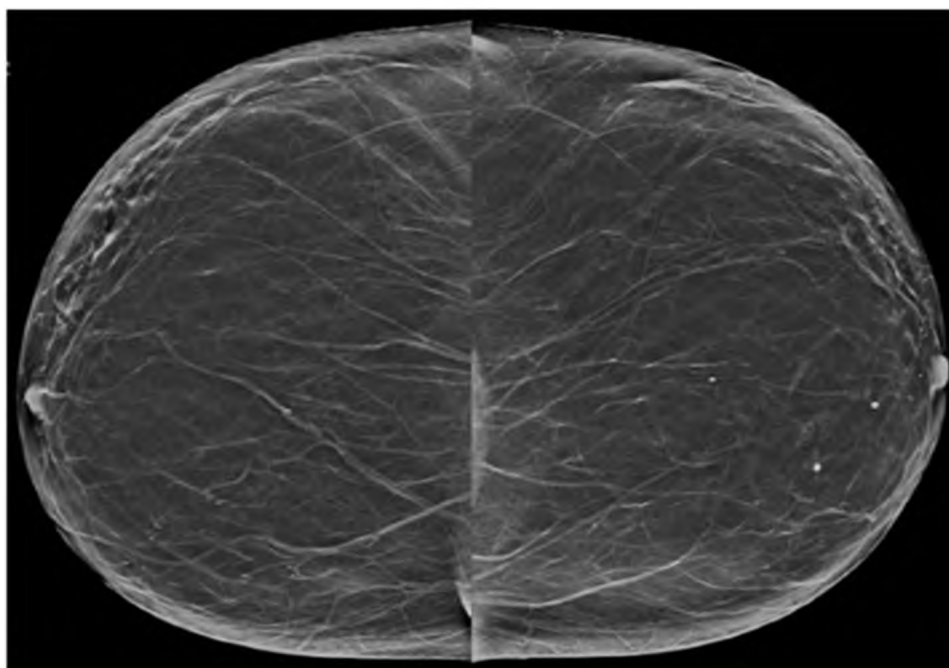


Figure 2 – Breast Density: BI-RADS breast composition category A, almost entirely fatty (SM).

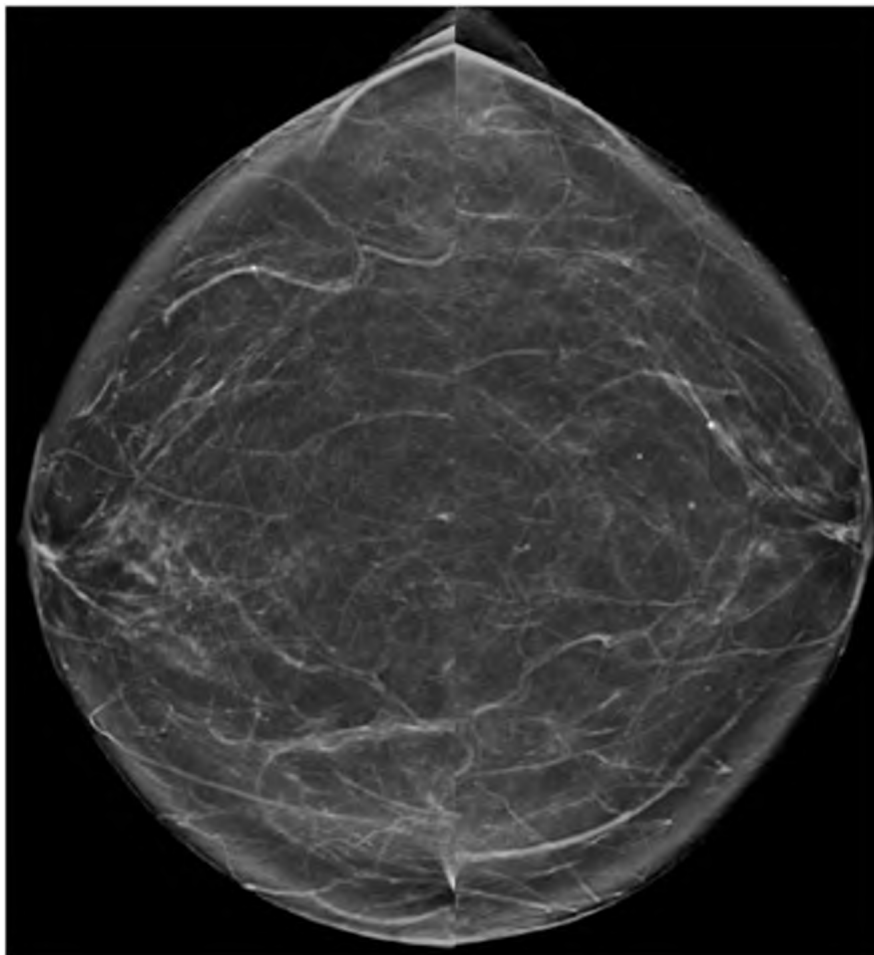


Figure 3 – Breast Density: BI-RADS breast composition category A, almost entirely fatty (SM).



Figure 4 – Breast Density: BI-RADS breast composition category A, almost entirely fatty (DM).

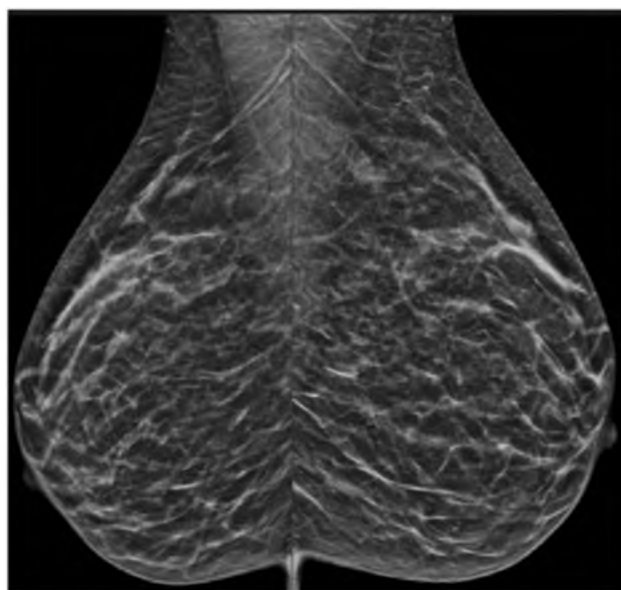


Figure 5 – Breast Density: BI-RADS breast composition category B, scattered areas of fibroglandular density (SM).

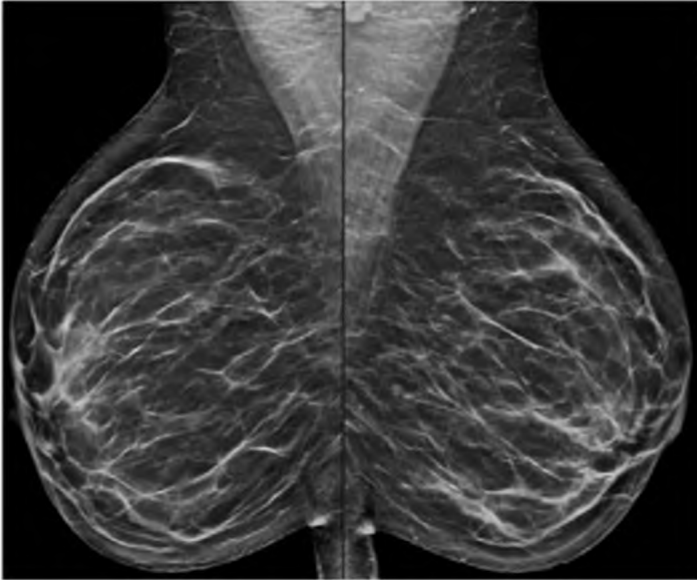


Figure 6 – Breast Density: BI-RADS breast composition category B, scattered areas of fibroglandular density (SM).



Figure 7 – Breast Density: BI-RADS breast composition category B, scattered areas of fibroglandular density (DM).

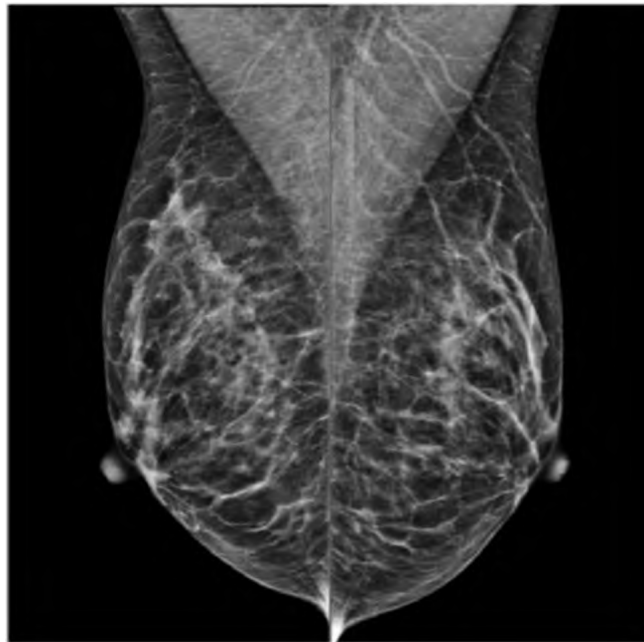


Figure 8 – Breast Density: BI-RADS breast composition category B, scattered areas of fibroglandular density (DM).

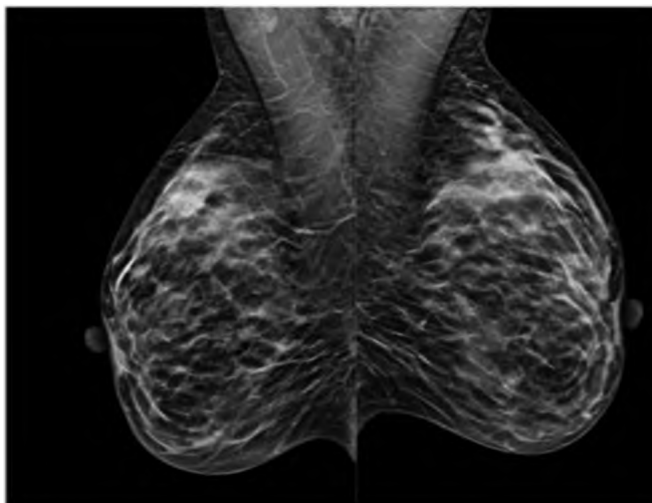


Figure 9 – Breast Density: BI-RADS breast composition category C, heterogeneously dense, which may obscure small masses (DM).

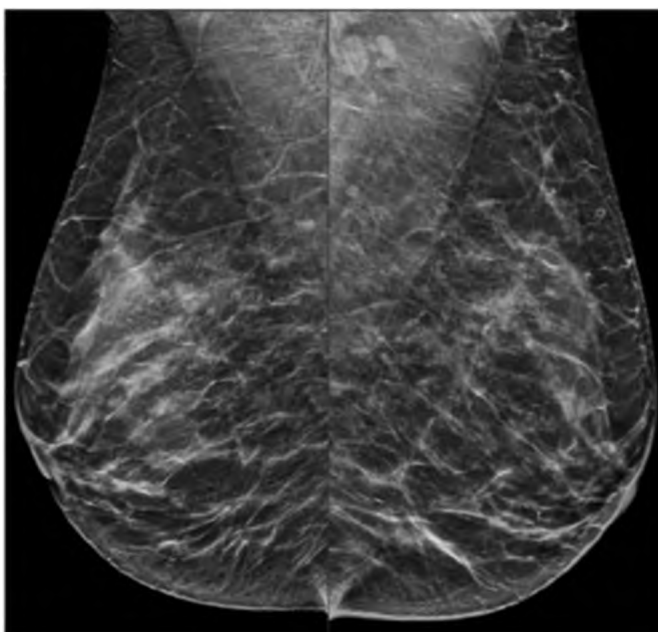


Figure 10 – Breast Density: BI-RADS breast composition category C, heterogeneously dense, which may obscure small masses (SM). Note that although the overall volume of dense tissue is not large, a cancer could be obscured in areas containing dense breast tissue.

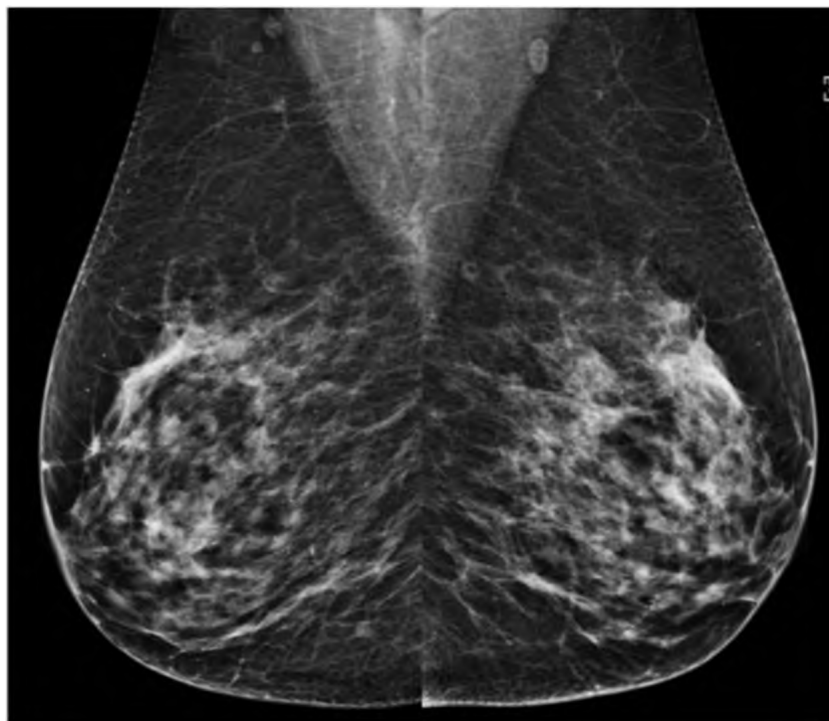


Figure 11 – Breast Density: BI-RADS breast composition category C, heterogeneously dense, which may obscure small masses (DM).

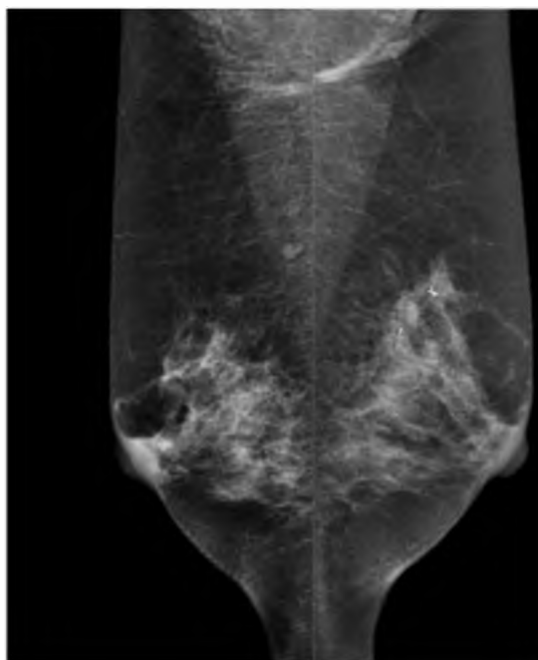


Figure 12 – Breast Density: BI-RADS breast composition category C, heterogeneously dense, which may obscure small masses (DM).

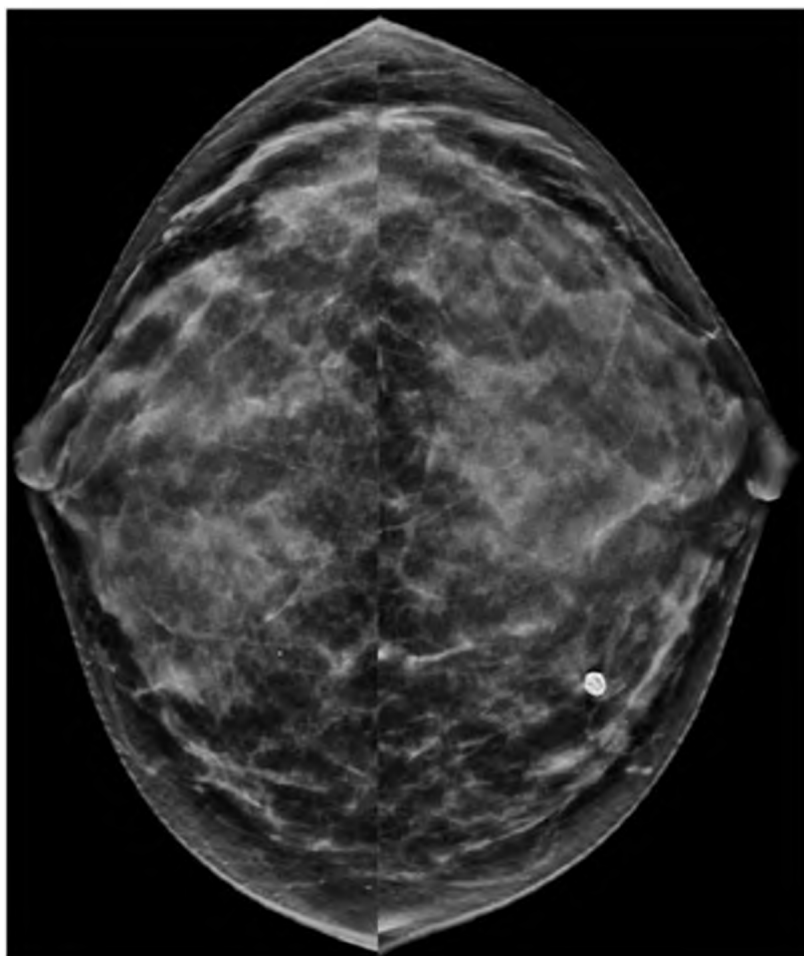


Figure 13 – Breast Density: BI-RADS breast composition category D, extremely dense, which lowers the sensitivity of mammography (SM).

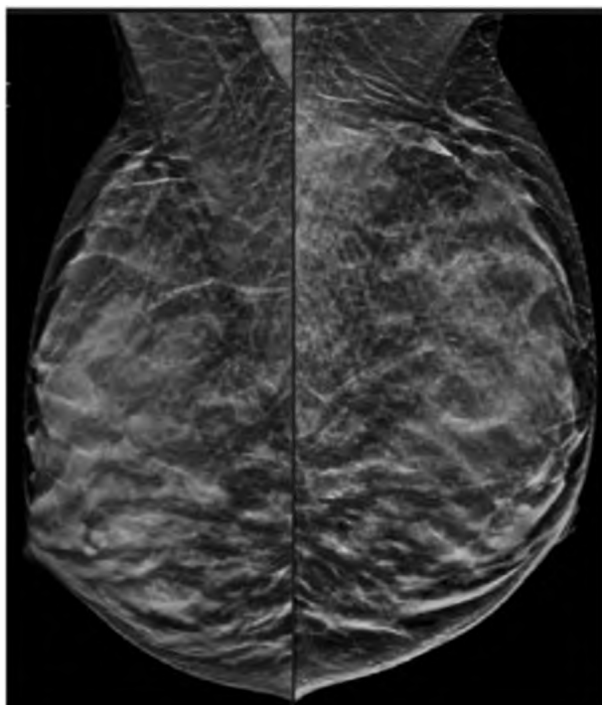


Figure 14 – Breast Density: BI-RADS breast composition category D, extremely dense, which lowers the sensitivity of mammography (SM).



Figure 15 – Breast Density: BI-RADS breast composition category D, extremely dense, which lowers the sensitivity of mammography (SM).



Figure 16 – Breast Density: BI- RADS breast composition category D, extremely dense, which lowers the sensitivity of mammography (SM).